S18 ETP Crosswalk – Mapping to OSCAL Assessment Results

Purpose

The HIMS-CDI Evidence Traceability Package (ETP) is implemented as a machine-readable JSON schema that captures evidence artefacts, timestamps, governance outcomes, and regulatory mappings. This supplementary section presents a crosswalk between ETP fields and the NIST OSCAL (Open Security Controls Assessment Language) Assessment Results model to facilitate interoperability with Governance, Risk, and Compliance (GRC) systems.

Mapping Table

ETP Field (HIMS-CDI)	OSCAL Assessment Results Property	Notes
trace_id	observation.uuid	Unique identifier for the evidence record
model_id	observation.subject-reference	Reference to assessed system or model
timestamp	observation.collected	ISO 8601 assessment timestamp
clause_id	observation.implemented-requirements[].control-id	Regulatory clause identifier
evidence_hash	observation.remarks	SHA-256 evidence artefact hash
result (PASS/FAIL)	observation.result	Compliance outcome
cdi_score	Custom OSCAL extension	HIMS-CDI composite deployment score
governance_gates	observation.additional-context	Governance gate statuses
mitre_technique	observation.threat-id	MITRE ATT&CK technique identifier

Example ETP JSON Snippet


```
{
 "trace_id": "trace_2025_0421_001",
 "model_id": "HIMS-CDI_cal",
 "timestamp": "2025-04-21T14:32:10Z",
 "clause_id": "GDPR-32-1-d",
 "evidence_hash": "a3f5b8c...",
 "result": "PASS",
 "cdi_score": 0.77,
 "governance_gates": {
  "CDI_GE_0.70": true,
  "CCR_GE_0.80": true,
  "EDGE_FEASIBLE": true
 },
 "mitre_technique": "T1566.001"
}
```

Conclusion

The crosswalk demonstrates that the HIMS-CDI Evidence Traceability Package aligns structurally with the OSCAL Assessment Results model, thereby enabling straightforward integration with enterprise GRC platforms. Framework-specific extensions, including CDI scores and governance-gate states, can be accommodated using OSCAL's extensible property mechanisms.